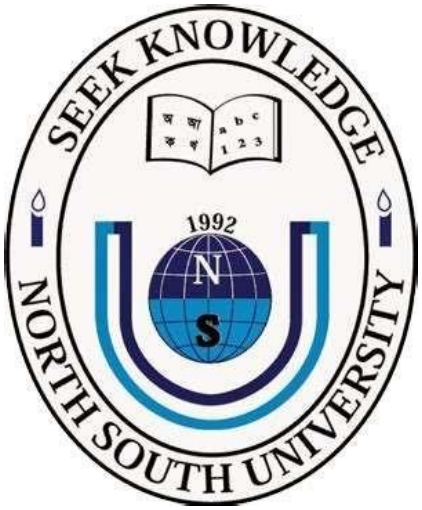
Analog Electronics II: Op-amp & 555 Timer



Course instructor:

Dr. Mohsin Sajjad

Assistant Professor

North South University

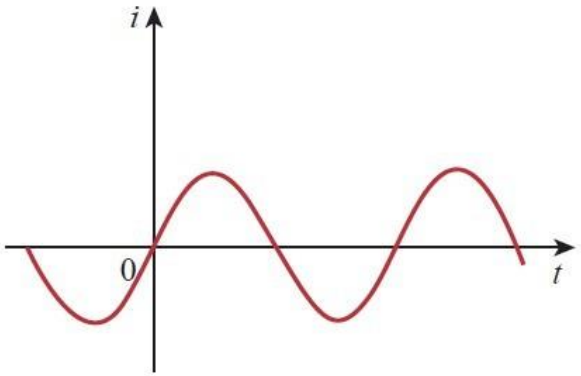
EEE 311

Lecture # 1

DC vs AC

A direct current (dc) flows only in one direction and can be constant or time varying.





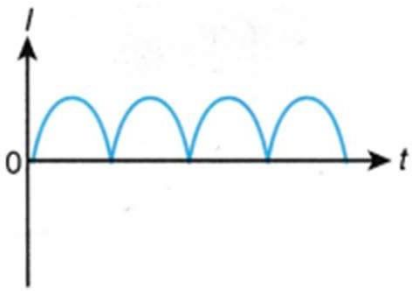
Source: Alexander and Sadiku

DC or AC?

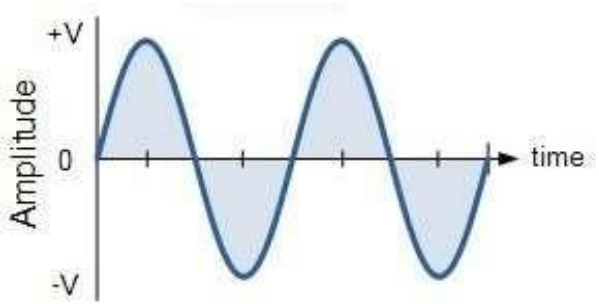
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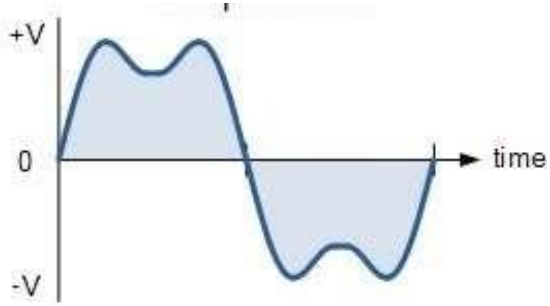
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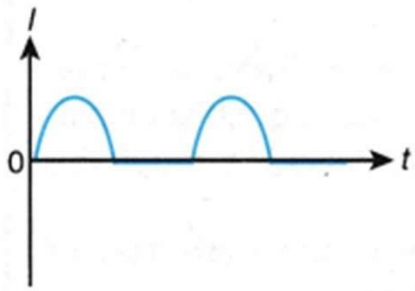
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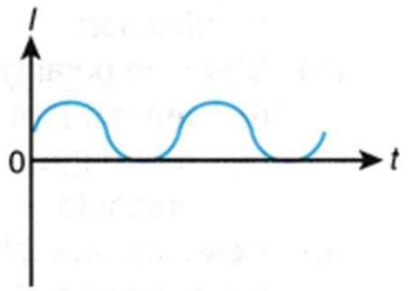
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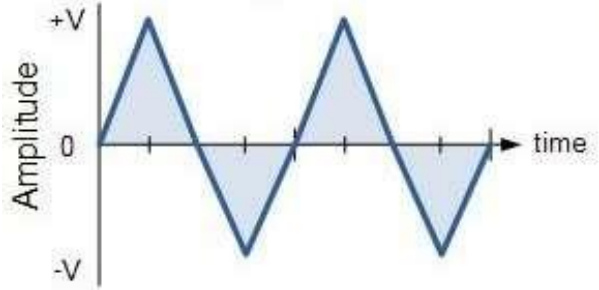
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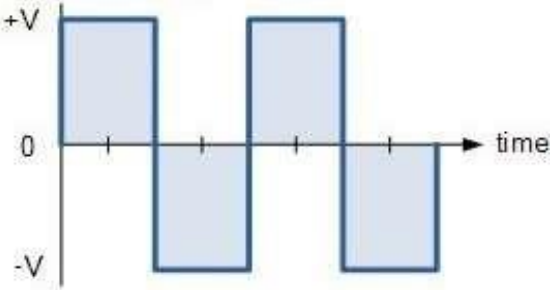
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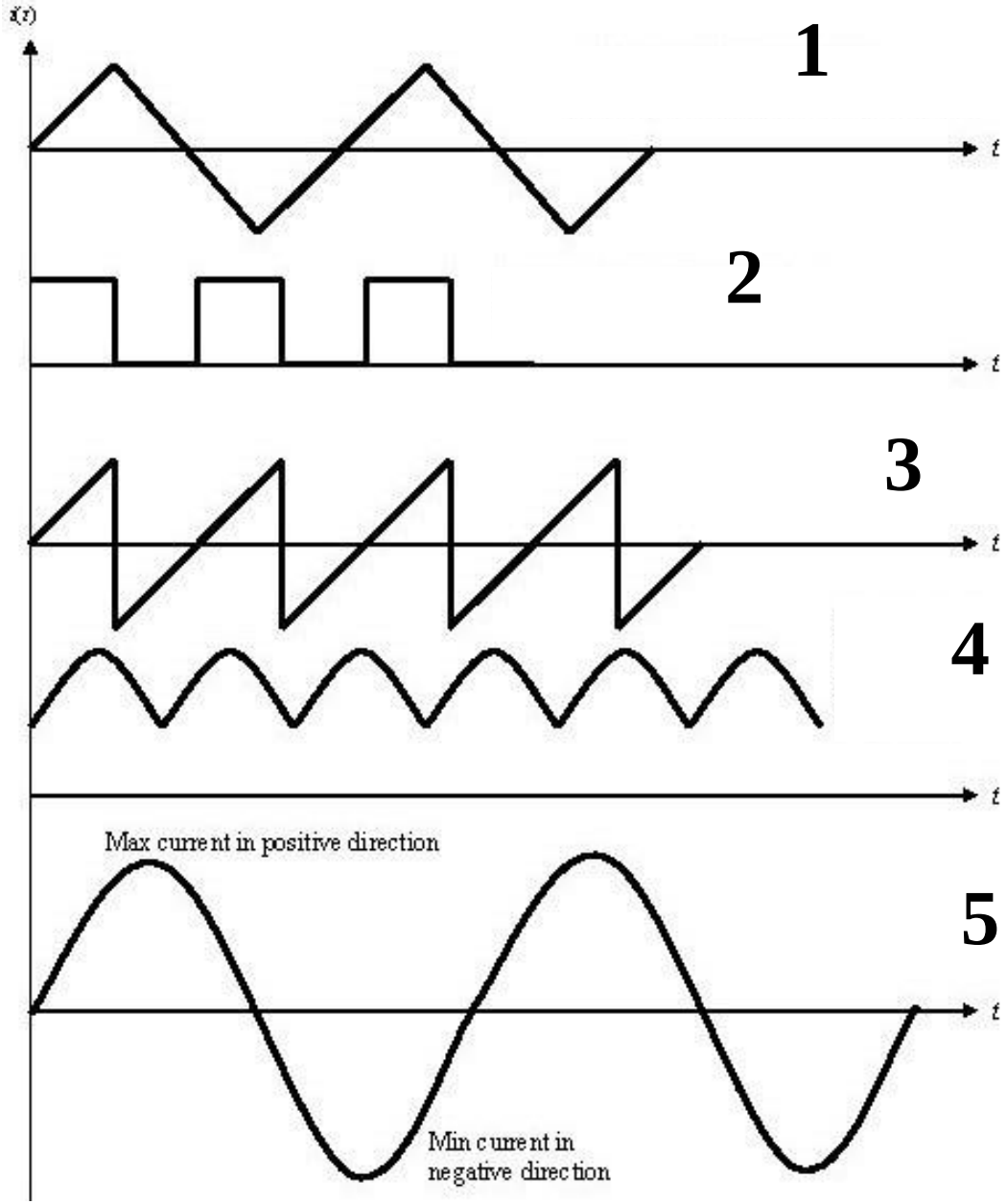
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8



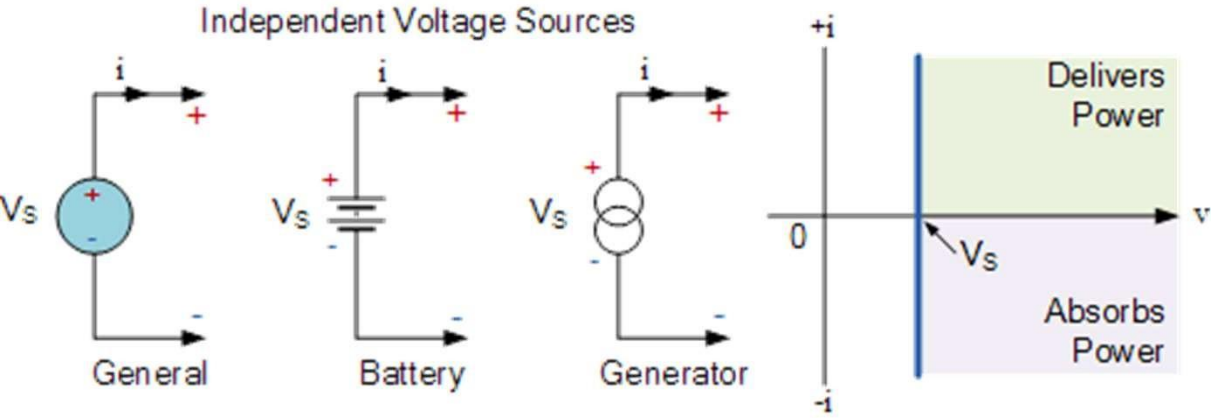
DC or AC?


x(t)
1
t
2
t
3
t
4
t
Max current in positive direction
5
t
Min current in
negative direction

Active circuit elements - independent

Generators, Batteries, Voltage and Current sources

An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.



ElectronicsTutorials

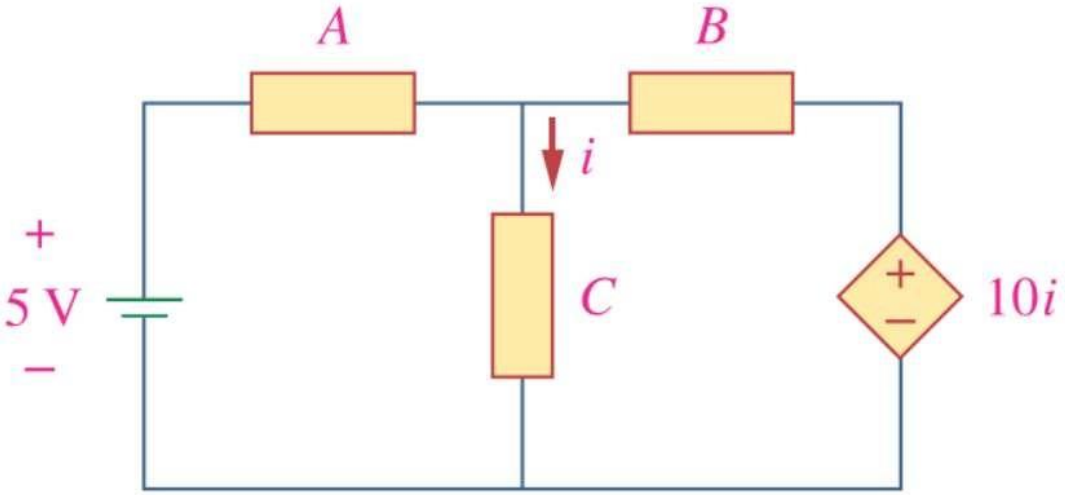
Source: Alexander and Sadiku

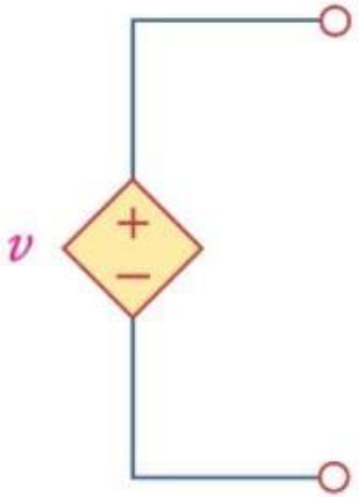
Active circuit elements - dependent

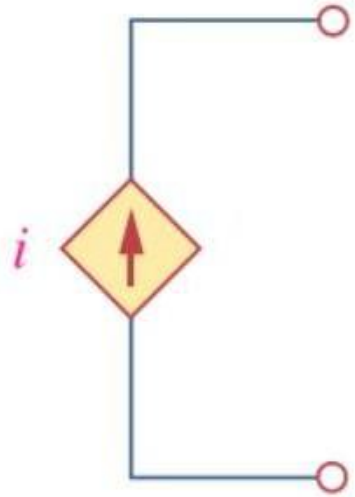
Voltage and Current sources

An **ideal dependent** (or **controlled**) **source** is an active element in which the source quantity is controlled by another voltage or current.

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).

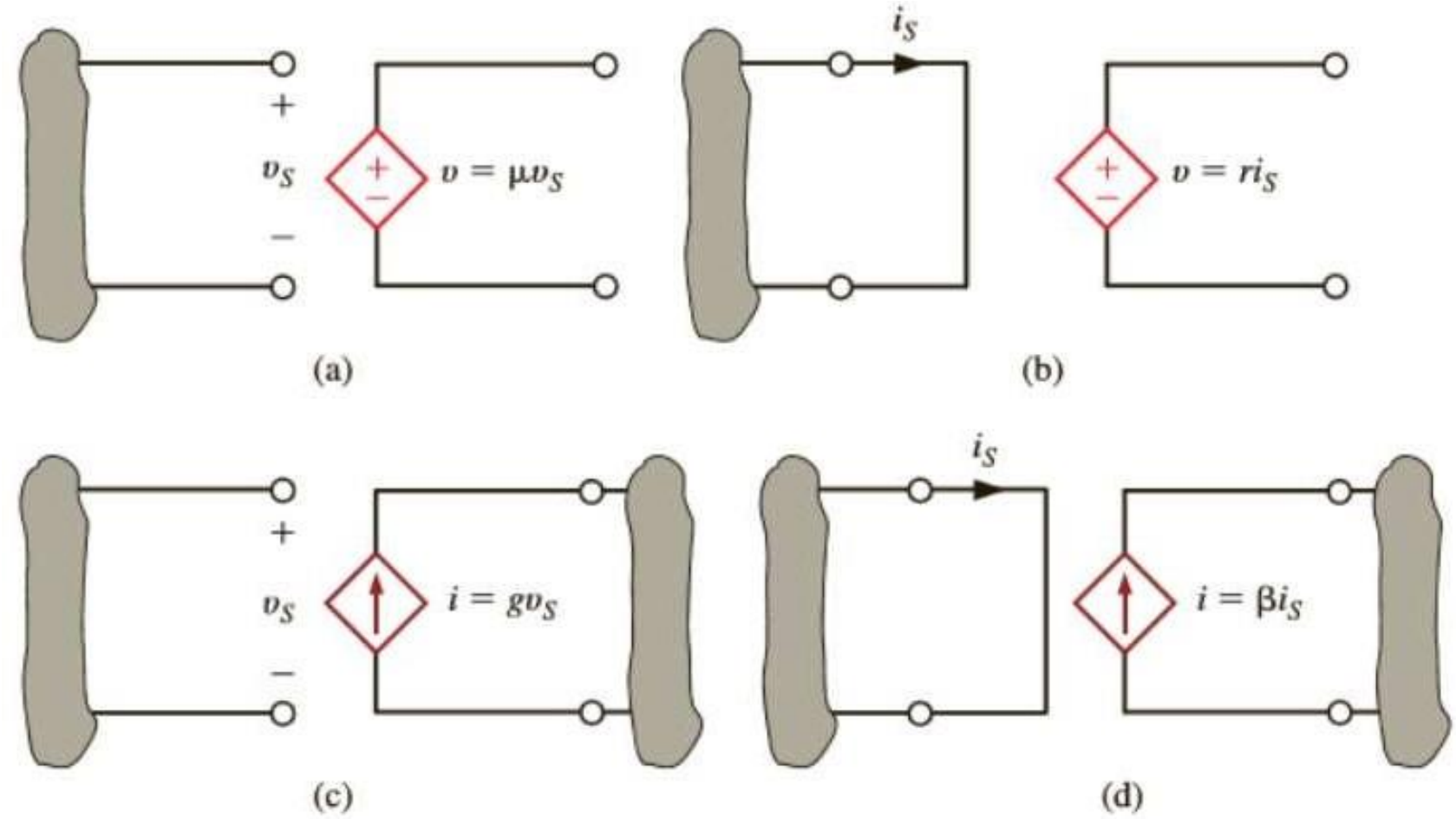






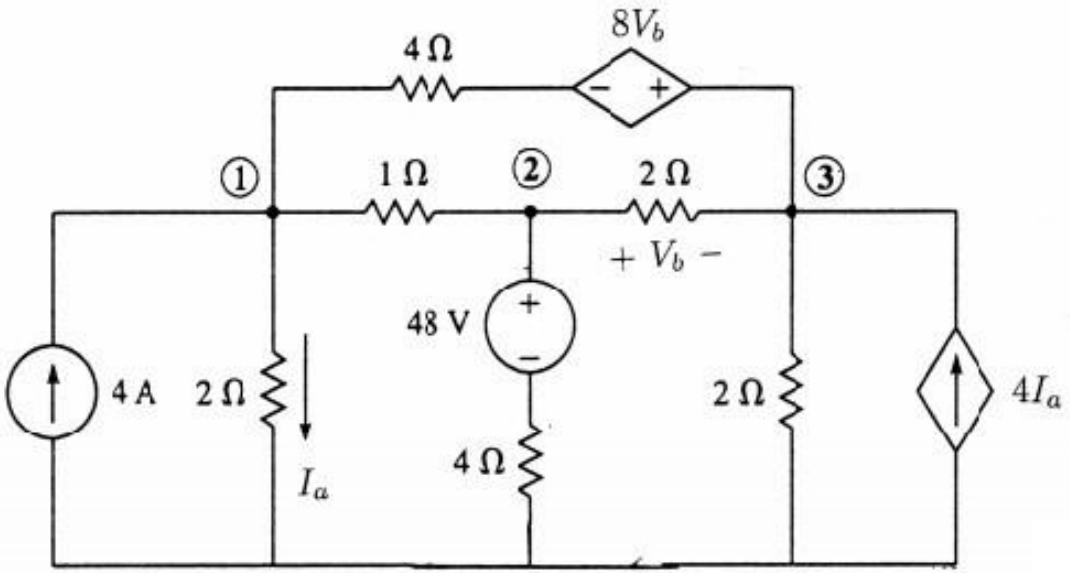
Source: Alexander and Sadiku

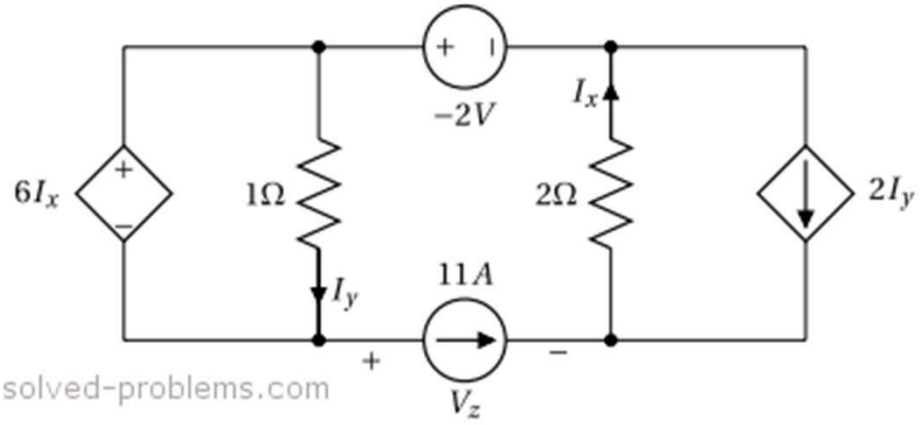
Active circuit elements - dependent

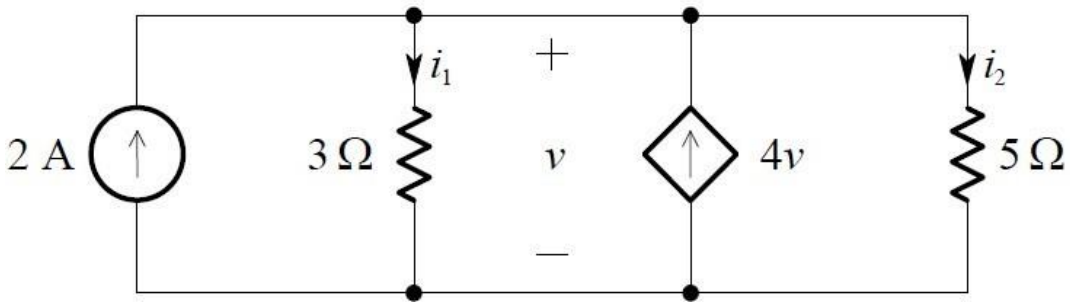


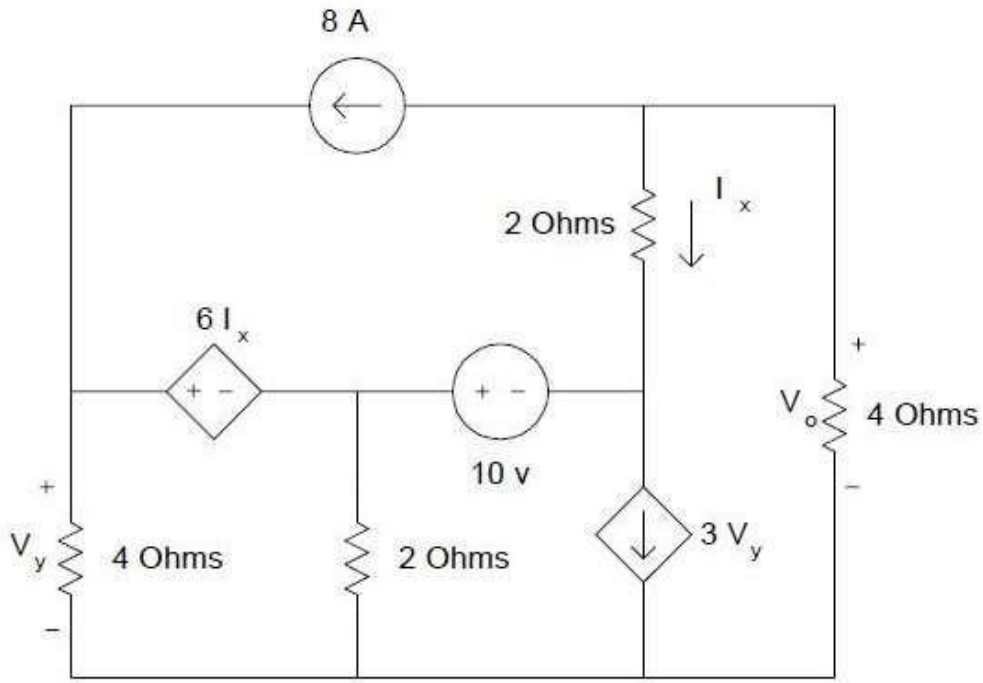
Source: Irwin and Nelms

Which type of Dependent source?



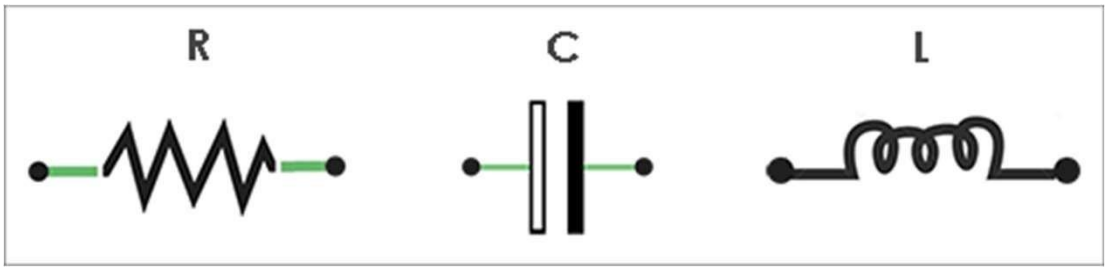






Source: Irwin and Nelms

Passive circuit elements

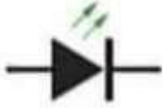
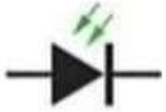
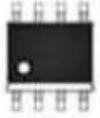
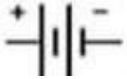


Resistor, Capacitor, Inductor

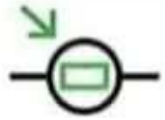
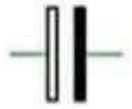
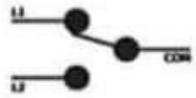
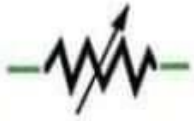
Passive elements are not capable to provide energy on their own

Source: https://www.codrey.com/dc-circuits/active-and-passive-elements/

ACTIVE

Transistor		
Diode		
LED		
Photodiode		
Integrated Circuit		-
Operational Amplifier		
Seven Segment Display		
Battery		

PASSIVE

Resistor		
LDR		
Thermistor		
Capacitor		
Inductor		
Switch		
Variable Resistor		
Transformer		

HACKATRONIC

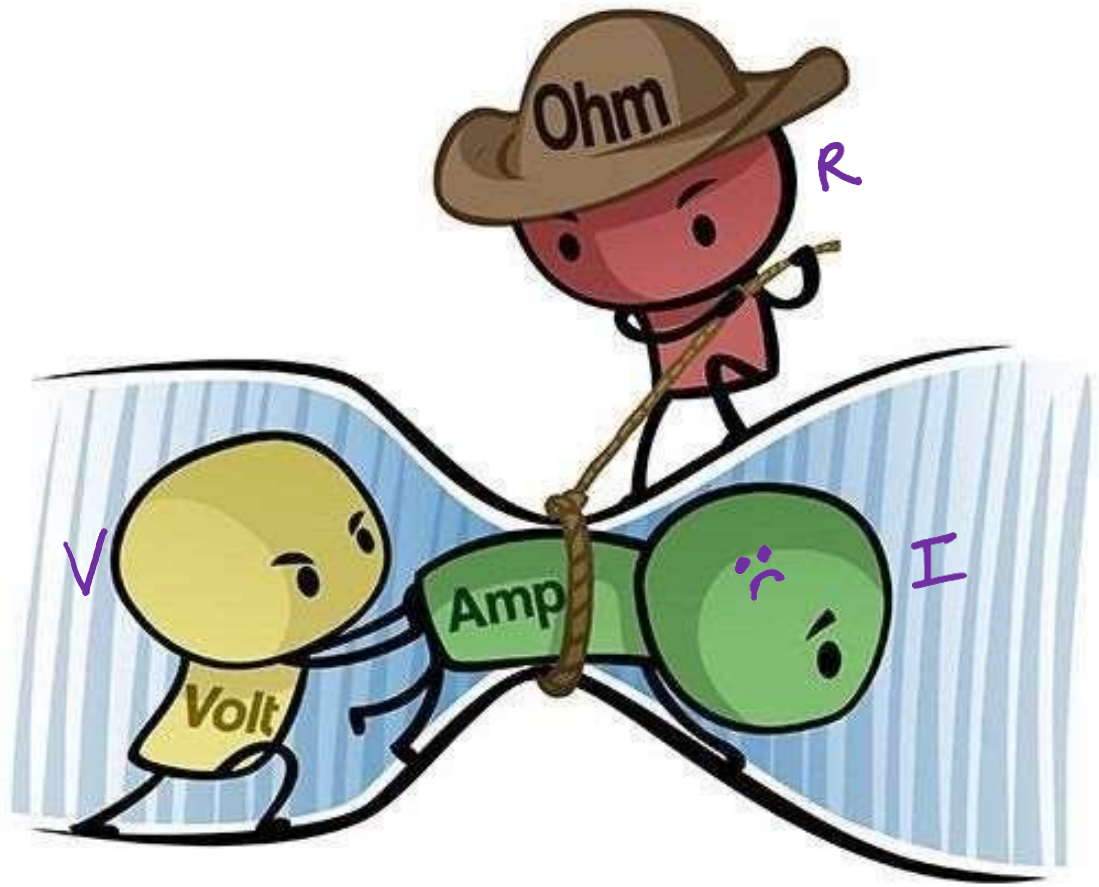
BASIS	ACTIVE COMPONENTS	PASSIVE COMPONENT
Nature of source	Active components deliver power or energy to the circuit.	Passive elements utilizes power or energy in the circuit.
Examples	Diodes, Transistors, SCR, Integrated circuits etc.	Resistor, Capacitor, Inductor etc.
Function of the component	Devices which produce energy in the form of voltage or current.	Devices which stores energy in the form of voltage or current.
Power Gain	They are capable of providing power gain.	They are incapable of providing power gain.

BASIS	ACTIVE COMPONENTS	PASSIVE COMPONENT
Flow of current	Active components can control the flow of current.	Passive components cannot control the flow of the current.
Requirement of external source	They require an external source for the operations.	They do not require any external source for the operations.
Nature of energy	Active components are energy donor.	Passive components are energy acceptor.

HACKATRONIC

Current, voltage, resistance

Ohm's law



$I \propto \frac{1}{R}$ (V const)

$I \propto V$ (R const)

Source: Amino

Ohm's law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

That is,

$$v \propto i \tag{2.2}$$

$$\boxed{v = iR}$$

The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

We may deduce from Eq. (2.3) that

$$R = \frac{v}{i} \tag{2.4}$$

so that

$$1\ \Omega = 1\ \text{V/A}$$

Source: Alexander and Sadiku

Example 2.1: An electric iron draws 2 A at 120 V. Find its resistance.

v = 120V

i = 2A

$R = \frac{v}{i} = \frac{120}{2} = 60\ \Omega$

Source: Alexander and Sadiku

2.2 Short Circuit

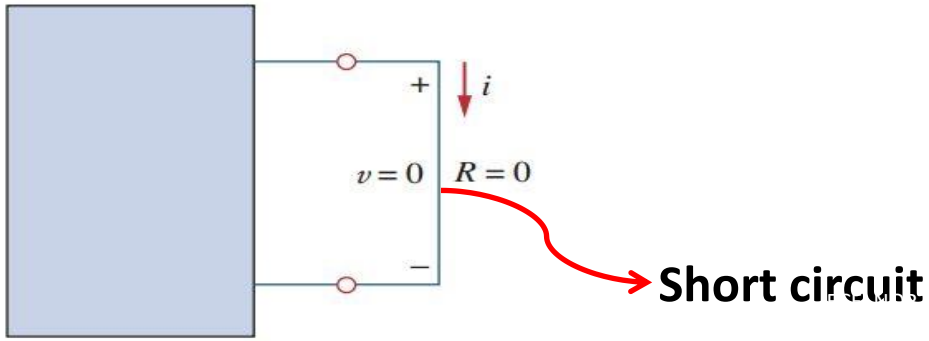
Short circuit: An element with $\mathrm{R} = 0\Omega$

Short circuit: circuit element with resistance approaching zero.

$$v = iR = i \times 0 = 0V$$

Voltage is zero, current is not zero

Usually, a connecting wire assumed to be a perfect conductor



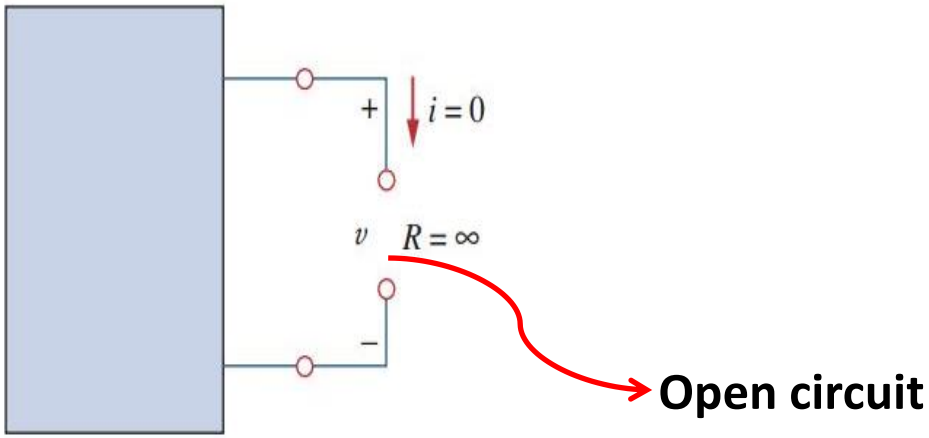
2.2 Open Circuit

Open circuit: An element with $\mathrm{R} = \infty\Omega$

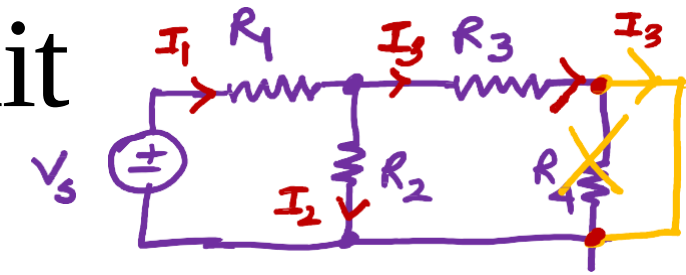
Open circuit: circuit element with resistance approaching infinity.

$$i = \frac{v}{R} = \frac{v}{\infty} = v \times 0 = 0A$$

Current is zero, voltage is not zero



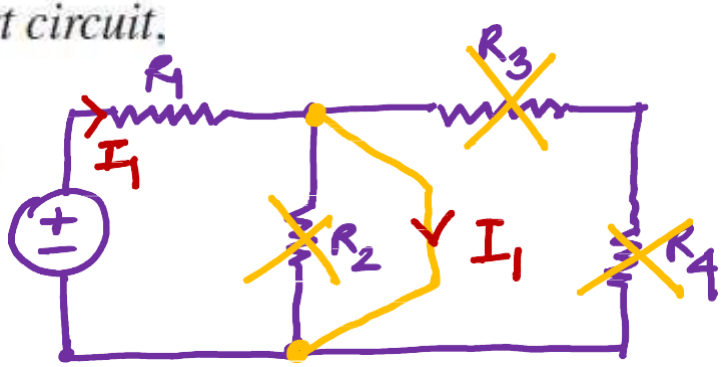
Short circuit and open circuit



A short circuit is a circuit element with resistance approaching zero.

$R = 0$ is called a *short circuit*.

$$v = iR = 0$$



$R = 0\ \Omega$

$V = 0\ V$

I_{sc}

↳ depends on the rest of network

An open circuit is a circuit element with resistance approaching infinity.

$R = \infty$ is known as an *open circuit*.

$R = \infty\ \Omega$

$I = 0\ A$

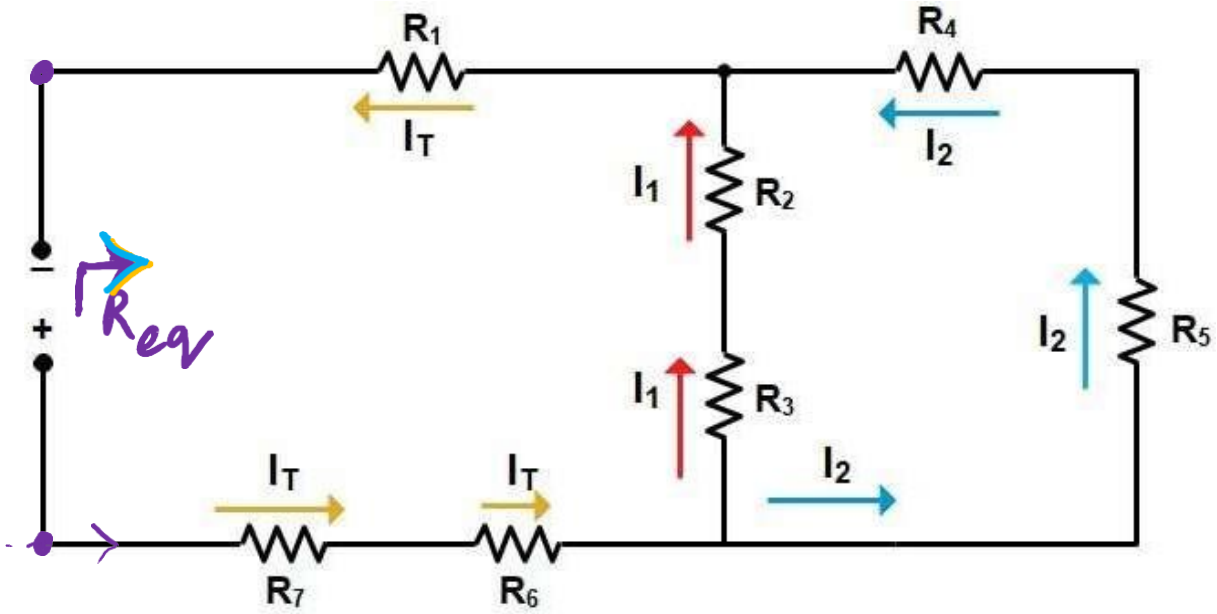
$$i = \lim_{R \to \infty} \frac{v}{R} = 0$$

$+\ V_{oc}\ -$

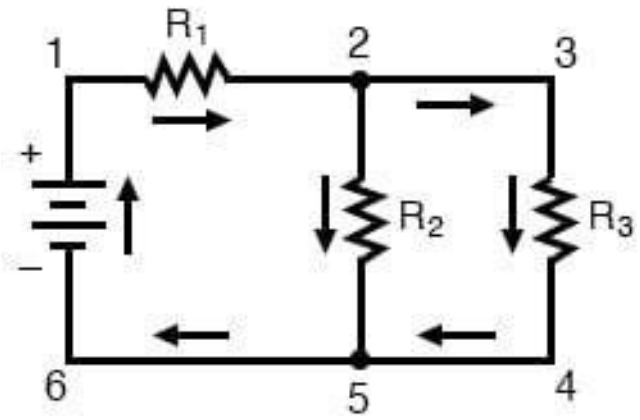
↳ depends on the rest of network

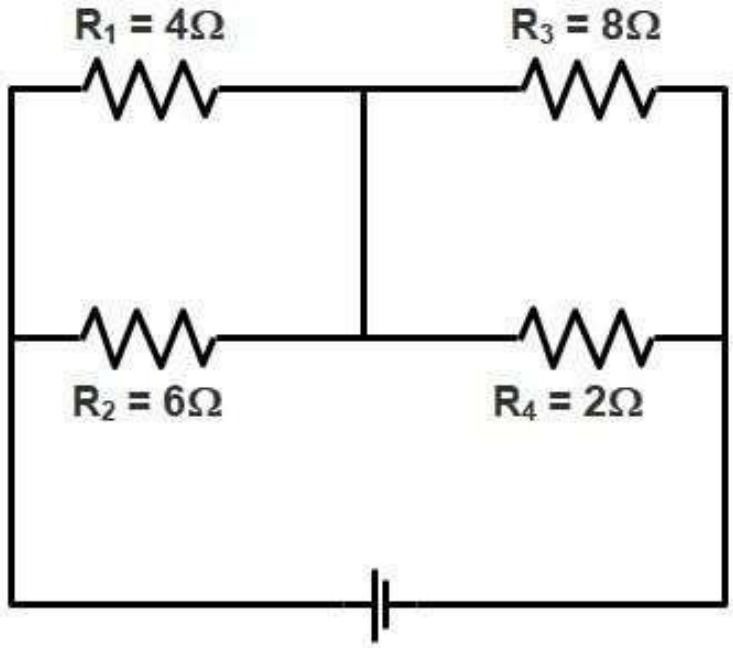
Source: Alexander and Sadiku

Series or parallel or none?



Tracing common current paths to identify series circuit





$R_{p_1} + R_{p_2}$

Source: Alexander and Sadiku

Example 2.9: Find R_{eq} for the circuit

Always start from the opposite side of R_{eq}

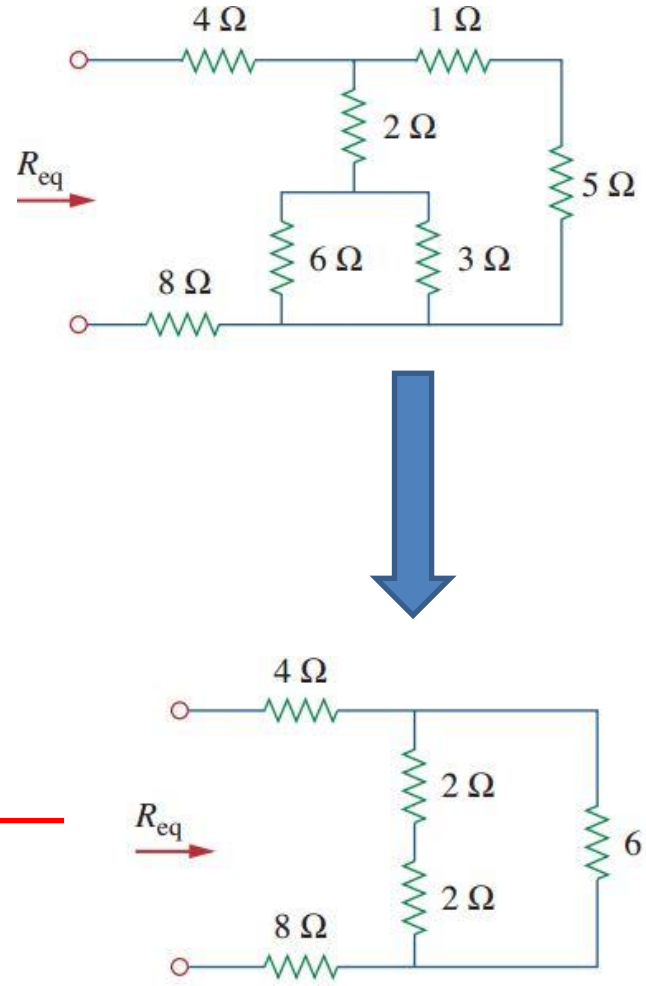
$1\ \Omega$ & $5\ \Omega$ series = $6\ \Omega$

$3\ \Omega$ & $6\ \Omega$ parallel = $\frac{3\times6}{3+6} = 2\ \Omega$

$2\ \Omega$ & $2\ \Omega$ series = $4\ \Omega$

$4\ \Omega$ & $6\ \Omega$ parallel = $\frac{4\times6}{4+6} = 2.4\ \Omega$

$4\ \Omega, 8\Omega$ & $2.4\ \Omega$ series = $14.4\ \Omega = R_{eq}$



2.4 Kirchhoff's current law (KCL)

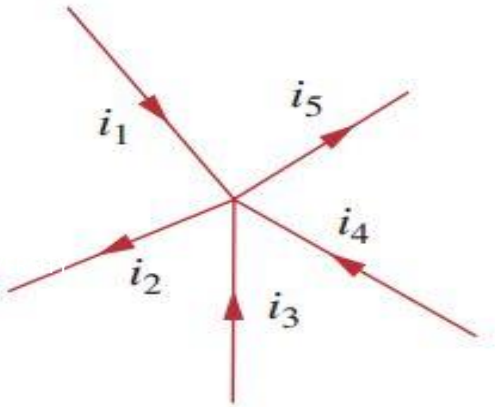
KCL: algebraic sum of currents entering a node (or a closed boundary) is zero.

Sum of the currents entering a node is equal to the sum of the currents leaving the node.

$$\sum_{node} i = 0A \rightarrow \sum_{node} i_{entering} = \sum_{node} i_{leaving}$$

$$+i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

$$\rightarrow \boldsymbol{i_1 + i_3 + i_4 = i_2 + i_5}$$



2.4 Kirchhoff's voltage law (KVL)

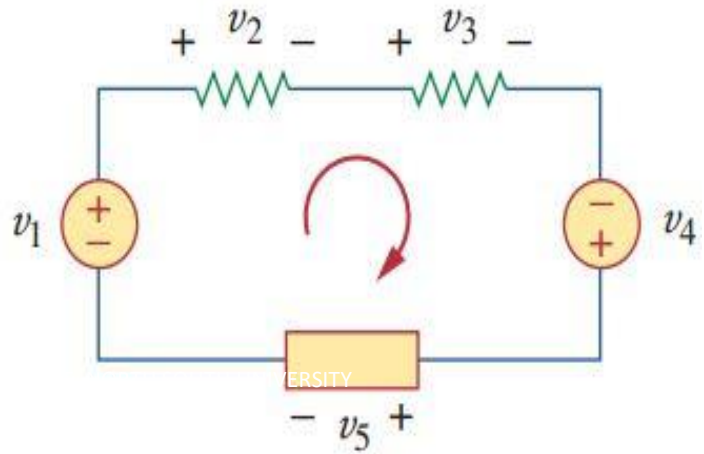
KVL: algebraic sum of all volt-ages around a closed path (or loop) is zero

Sum of voltage drops = Sum of voltage rises

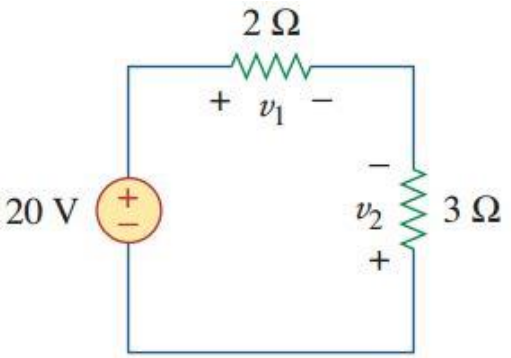
$$\sum_{loop} v = 0V \rightarrow \sum_{loop} v_{rise} = \sum_{loop} v_{drop}$$

$$-v_1 + v_2 + v_3 + v_5 - v_4 = 0$$

$$v_1 + v_4 = v_2 + v_3 + v_5$$



Example 2.5: Find v_1 and v_2

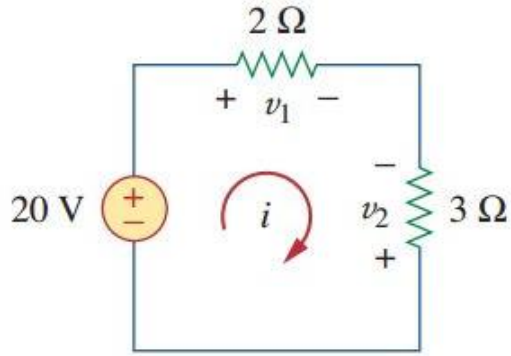


From Ohm's law,

$$v_1 = 2i, \qquad v_2 = -3i \tag{2}$$

Applying KVL around the loop gives

$$+20 + v_1 - v_2 = 0 \tag{2}$$



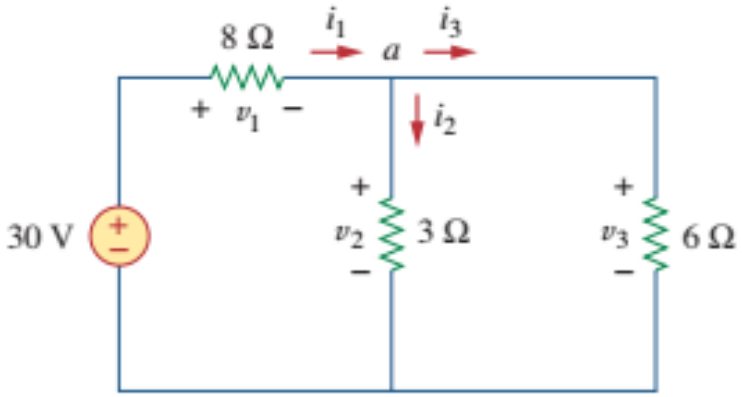
Substituting Eq. (2.5.1) into Eq. (2.5.2), we obtain

$$-20 + 2i + 3i = 0 \quad \text{or} \quad 5i = 20 \quad \Rightarrow \quad i = 4\ \text{A}$$

Substituting i in Eq. (2.5.1) finally gives

$$v_1 = 8\ \text{V}, \qquad v_2 = -12\ \text{V}$$

Example 2.8: Find currents and voltages in the circuit



- 6 unknowns. Need 6 equations.
- Ohm's Law: $v_1 = 8i_1; v_2 = 3i_2; v_3 = 6i_3$
- 3Ω and 6Ω: parallel
- KVL Loop 2: $v_2 = 3i_2 = v_3 = 6i_3$
- KCL at node a: $i_1 = i_2 + i_3$
- KVL Loop 1: $-30V + v_1 + v_2 = 0$
- $i_1 = 3A; i_2 = 2A; i_3 = 1A$
- $v_1 = 24V; v_2 = 6V; v_3 = 6V$

